

AI-Powered Smart Shopping Cart with Intelligent Shelf Monitoring and Real-Time Inventory Management

1. Project Overview

This project proposes an **AI-powered smart shopping system** that combines **Artificial Intelligence (AI), Computer Vision, IoT, RFID, weight sensors, and a mobile application** to create a seamless shopping experience with automatic billing and real-time inventory management.

The objective is to eliminate traditional checkout queues while maintaining accurate inventory tracking. Customers simply pick products, scan them (or have them automatically verified), place them in the smart cart, and complete payment through a mobile application.

Unlike expensive fully camera-based systems such as Amazon Go, this solution uses a **hybrid verification approach**, combining:

- AI-based shelf monitoring
- Weight validation
- IoT-enabled smart carts
- Real-time inventory synchronization

This significantly improves reliability while keeping the prototype affordable.

2. Problem Statement

Traditional supermarkets face several challenges:

- Long billing queues
- Manual barcode scanning
- Inventory inaccuracies
- Delayed stock updates
- Product misplacement
- Shoplifting and fraud
- High manpower requirements

This project addresses these issues through an integrated smart shopping ecosystem.

3. Objectives

- Reduce checkout time
 - Eliminate billing queues
 - Track products in real time
 - Automatically update inventory
 - Detect product movement using AI
 - Validate purchases using RFID and weight sensors
 - Improve overall supermarket operations
 - Build a scalable architecture for future expansion
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4. Prototype Scope

The prototype simulates a miniature supermarket.

Store Setup

- 2 Smart Shelves
- 2 Products
 - Shelf 1 → Milk
 - Shelf 2 → Biscuits
- 1 Smart Shopping Cart
- 1 AI Camera
- Customer Mobile Application
- Supermarket Management Dashboard
- Backend Server
- PostgreSQL Database

Although the prototype is small, the software architecture is designed to support:

- Hundreds of shelves
 - Thousands of products
 - Multiple shopping carts
 - Multiple stores
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5. System Components

A. Smart Shopping Cart

Hardware

- Arduino Uno
- ESP8266 / NodeMCU WiFi Module
- Barcode Scanner
- Load Cell
- HX711 Amplifier
- LCD Display (optional)
- Green LED
- Red LED
- Active Buzzer
- QR Code
- Battery/Power Supply

Responsibilities

- Pair with customer
 - Scan product barcode
 - Measure cart weight
 - Validate purchases
 - Display cart status
 - Send data to backend
 - Provide visual/audio feedback
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B. Smart Shelf

Each shelf contains:

- Fixed Product Type
- Shelf ID
- AI Camera Coverage
- Load Cell
- HX711 Module

Example

Shelf 1

Milk

Shelf 2

Biscuits

Responsibilities

- Detect product removal
 - Detect product return
 - Monitor stock quantity
 - Detect empty shelves
 - Detect incorrect product placement
 - Monitor shelf weight
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C. AI Vision System

The prototype uses a **single USB webcam** positioned to monitor both shelves.

Computer Vision performs:

- Person detection
- Product detection
- Shelf monitoring
- Product movement tracking
- Product return detection
- Wrong shelf detection
- Customer-product association

Technologies:

- YOLO
 - OpenCV
 - Gemini 2.5 Flash (for advanced vision reasoning)
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D. Mobile Application

Customers can:

- Login
 - Pair cart using QR code
 - View shopping cart
 - Receive live bill updates
 - Remove products
 - Checkout
 - Make payments
 - View purchase history
 - Download invoices
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E. Supermarket Management Dashboard

Staff dashboard includes:

Product Management

- Add products
- Update products
- Delete products

Inventory Management

Track inventory across:

- Warehouse
- Shelf
- Shopping Cart
- Sold Products

Shelf Monitoring

- Shelf stock
- Camera status
- Shelf occupancy
- Empty shelf alerts

Cart Monitoring

- Connected carts
- Cart contents
- Customer sessions
- Checkout status

Analytics

- Daily sales
- Product popularity
- Inventory trends
- Customer activity

Ticket Management

Automatically generate maintenance tickets for:

- Camera offline
 - Barcode failure
 - Weight mismatch
 - Low stock
 - Inventory mismatch
 - Sensor failure
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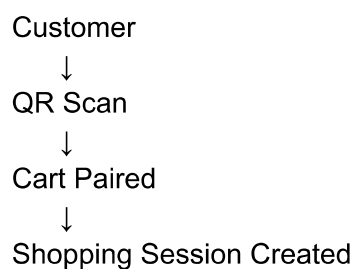
6. System Workflow

Step 1 – Customer Entry

Customer enters the supermarket.

Customer opens the mobile application.

Customer scans the QR code attached to a shopping cart.



Step 2 – Product Removed from Shelf

Customer picks a Milk packet.

Immediately:

- Shelf weight decreases.
- AI camera detects product removal.
- Vision system identifies the product.
- The event is linked to the active shopping session.

Example:

Shelf Weight

2000g

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1500g

Milk Removed

Backend records:

- Product removed
- Shelf ID
- Timestamp
- Customer session

Step 3 – Product Added to Cart

Customer scans the product barcode (prototype) or places the RFID-tagged product into the cart.

The cart performs:

- Barcode identification
- Weight validation

If successful:

- Green LED ON
- Buzzer beeps

- Product added to cart
 - Mobile app updates instantly
 - Inventory changes from **Shelf** → **Cart**
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Step 4 – Product Viewed and Returned

Customer picks a Biscuit packet.

After inspecting it:

Customer returns it to Shelf 2.

System detects:

- Shelf weight increases
- AI confirms return
- Inventory restored

No billing occurs.

Step 5 – Product Removed from Cart

Customer decides not to buy the Milk.

Customer removes it from the cart and places it back on Shelf 1.

System detects:

- Product removed from cart
- Cart weight decreases
- Shelf weight increases
- AI confirms return

Backend updates:

Cart → Shelf

Mobile app removes the item from the bill.

Step 6 – Checkout

Customer opens the mobile app.

The application displays:

- Product list
- Quantities
- Total amount
- Taxes (optional)

Customer pays digitally.

Receipt is generated automatically.

No billing counter is required.

7. Product Life Cycle

Warehouse



Shelf



Customer Picks Product



Shelf Detection



Cart Verification



Shopping Cart



Checkout



Sold Inventory

If returned:

Cart



Shelf

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Inventory Restored

8. Inventory Management

The system continuously tracks inventory across four locations:

Location	Description
Warehouse	Products in storage
Shelf	Products available for customers
Shopping Cart	Products currently selected
Purchased	Successfully sold products

Inventory updates automatically whenever a product changes state.

9. Hybrid Verification Logic

To improve accuracy, the system combines multiple verification methods.

Verification Layer	Purpose
AI Camera	Detects product movement and customer interaction
Shelf Weight	Confirms product removed or returned
Barcode Scan	Identifies the product (prototype)
RFID	Confirms product inside the cart
Cart Weight	Validates expected vs actual weight

A product is confirmed only when the required verification checks pass.

10. Fraud Detection

Case 1 – Product Picked but Not Added to Cart

Events:

- Shelf weight decreases
- AI detects removal
- No barcode/RFID detected

System response:

- Wait for a timeout
 - Display: "Please place the item in the cart."
 - Blink LED
 - Log the event
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Case 2 – Hidden Product

Expected cart weight does not match actual cart weight.

Actions:

- Red LED
 - Buzzer
 - Management notification
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Case 3 – Wrong Shelf Placement

Customer places Milk on Biscuit shelf.

AI detects:

- Incorrect product
- Wrong shelf

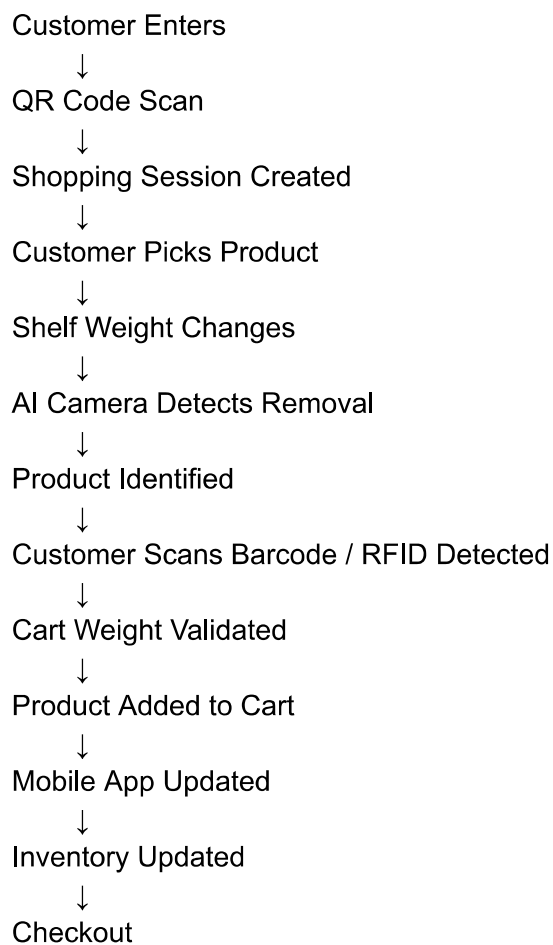
Inventory remains unchanged until corrected.

Case 4 – Multiple Customers in Same Zone (Prototype Limitation)

If the camera detects multiple customers near the same shelf:

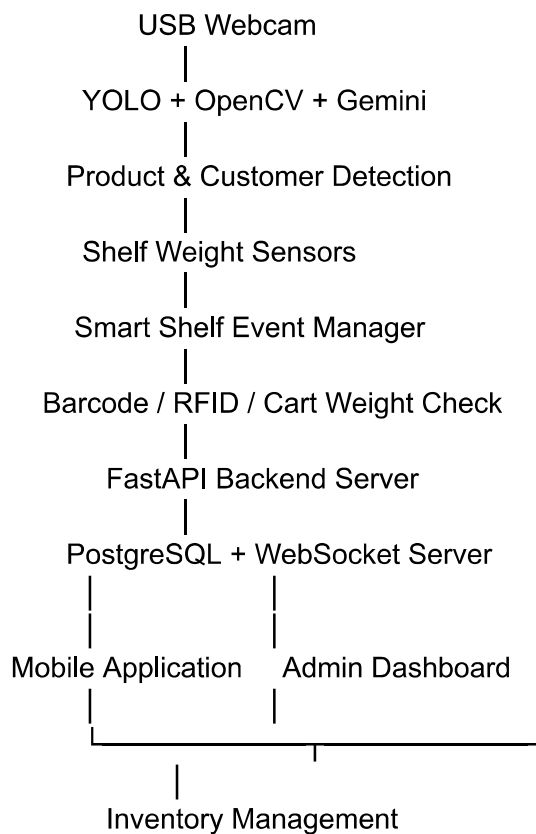
- Pause automatic association
 - Display a warning
 - Request one customer to pick items at a time
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11. Complete System Flow



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Digital Payment
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Invoice Generated

12. Overall System Architecture



13. Technologies Used

Artificial Intelligence

- YOLO
- OpenCV
- Gemini 2.5 Flash

Backend

- Python 3.11
- FastAPI
- WebSockets

Database

- PostgreSQL

Frontend

- React (Admin Dashboard)
- Flutter or React Native (Mobile App)

Hardware

- Arduino Uno
 - ESP8266 / NodeMCU
 - MFRC522 RFID Reader
 - Barcode Scanner
 - Load Cell
 - HX711
 - USB Webcam
 - LCD Display
 - LEDs
 - Buzzer
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14. Hardware Requirements

Component	Quantity
Arduino Uno	1
NodeMCU ESP8266	1
MFRC522 RFID Reader	1
RFID Tags	10

Barcode Scanner	1
USB Webcam (720p)	1
Load Cells	3
HX711 Modules	3
LCD Display	1
Active Buzzer	1
LEDs	10
Power Supply	1
Breadboards & Wires	As required

Estimated prototype cost: ₹3,500–₹4,500

15. Key Features

Customer Features

- QR-based cart pairing
- Real-time shopping cart
- Live bill updates
- Digital checkout
- Purchase history
- Invoice download

Smart Cart Features

- Barcode scanning
- RFID verification
- Weight validation
- LCD display
- Wi-Fi connectivity
- LED and buzzer feedback

AI Shelf Features

- Product removal detection
- Product return detection
- Shelf stock monitoring
- Empty shelf alerts
- Wrong product placement detection

Inventory Features

- Warehouse tracking
- Shelf tracking
- Cart tracking
- Sold inventory tracking
- Automatic stock synchronization

Management Features

- Product management
 - Inventory management
 - Live shelf monitoring
 - Cart monitoring
 - Notifications
 - Analytics dashboard
 - Automatic ticket generation
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16. Advantages

- Faster shopping experience
 - Eliminates checkout queues
 - Automatic billing
 - Real-time inventory updates
 - Reduced manual work
 - Improved stock accuracy
 - Fraud detection through multi-layer verification
 - Cost-effective compared to fully vision-based stores
 - Modular and scalable architecture
 - Easy expansion to multiple stores
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17. Future Enhancements

- Support multiple shopping carts
 - Support hundreds of shelves and products
 - AI-powered personalized recommendations
 - Online payment gateway integration
 - Cloud deployment
 - Multi-store management
 - Digital twins for inventory
 - Predictive stock replenishment
 - Customer loyalty and rewards
 - Advanced analytics and demand forecasting
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18. Conclusion

The **AI-Powered Smart Shopping Cart with Intelligent Shelf Monitoring and Real-Time Inventory Management** integrates AI, IoT, RFID, weight sensing, computer vision, and mobile technologies into a unified retail solution. By combining **AI-based shelf observation** with **RFID, barcode scanning, and weight validation**, the system delivers reliable automatic billing, real-time inventory synchronization, and fraud detection while remaining affordable for a prototype. Although demonstrated using **two smart shelves, two products, and one shopping cart**, the architecture is modular and scalable, making it suitable for deployment in larger supermarkets with multiple carts, products, and store locations.